EA Collaborative Learning - Personalized and Private Peer-to-Peer Machine Learning

Louis Duvignacq Eloïse Trabut Quentin Nguyen Alexandre Jiao

Abstract

As connected personal devices proliferate, there is increasing demand for machine learning paradigms that leverage distributed data while maintaining strict privacy guarantees. Personalized and Private Peer-to-Peer Machine Learning, originally proposed by (Bellet et al., 2018), offers a decentralized coordinate descent framework enhanced with differential privacy. In this paper, we critically analyze the assumptions underpinning this method, including reliance on strong convexity assumptions, trust models based on honest-but-curious agents, and the construction of similarity weights. We propose several extensions addressing nonconvex loss functions, adversarial behavior, and graph configurations.

1. Introduction

Decentralized learning has emerged as a key paradigm for privacy-sensitive machine learning applications where data cannot be centralized. Classic distributed learning architectures typically rely on server-based coordination (Nedic & Ozdaglar, 2009; Duchi et al., 2012), but recent interest has shifted toward *fully* peer-to-peer (P2P) systems, motivated by scalability and privacy concerns.

Bellet et al. (2018) propose a decentralized multi-task learning framework in which each agent maintains a personalized model, communicates only with its graph neighbors, and ensures differential privacy of its local dataset. Their algorithm based on an asynchronous block coordinate descent method, optimizes a graph-regularized objective encoding both local empirical risk and similarity between agents.

This paper provides a structured discussion of the assumptions that make the analysis in Bellet et al. (2018) tractable and highlights opportunities for extending the underlying ideas. We examine assumptions on the convexity of the loss function, network topology, and adversarial capabilities, and discuss their implications for convergence, privacy, and practical deployment.

2. Related Work

Decentralized Optimization. Most work in fully decentralized learning has focused on the *distributed consensus problem*, where the goal is to find a single global model minimizing the sum of local losses (Nedic & Ozdaglar, 2009; Ram et al., 2010; Duchi et al., 2012; Colin et al., 2016). While effective for global tasks, these approaches do not address personalized objectives. Vanhaesebrouck et al. (2017) studied decentralized personalized learning using a gossip-based ADMM algorithm; however, their approach lacks privacy constraints and relies on auxiliary variables, leading to high communication costs. In contrast, Bellet et al. (2018) propose a block coordinate descent algorithm with linear convergence. Recent surveys highlight a shift toward secure decentralized frameworks that are resilient to single-point failures, moving beyond simple consensus to handle heterogeneous data (Liu et al., 2024).

Differential Privacy in Distributed Learning. Differential Privacy (DP) has traditionally relied on a "trusted curator" (centralized setting) (Chaudhuri et al., 2011) or local perturbation (Local DP). Local DP (Duchi et al., 2012; Kairouz et al., 2016) adds noise to data before release but often suffers from poor accuracy on small datasets. In the master/slave setting, algorithms typically aggregate locally trained models or perturb gradients (Pathak et al., 2010; Shokri & Shmatikov, 2015). Huang et al. (2015) considered a decentralized setting similar to ours but focused on a single global model with synchronous updates.

Recent advancements seek to overcome these limitations. For instance, Biswas et al. (2025) proposed *Zip-DL*, a framework that optimizes the utility-privacy trade-off in P2P settings without requiring the strict synchronization or global modeling assumptions of previous work.

Adversarial Attacks and Robustness. While Bellet et al. assume an honest-but-curious model, Hitaj et al. (2017) demonstrated that malicious participants in collaborative deep learning can use Generative Adversarial Networks

(GANs) to reconstruct sensitive class representatives, even under DP. Addressing such threats has become a priority in recent literature. Methods like "Remove-then-Clip" have been introduced to ensure Byzantine robustness against malicious updates (Yang & Ghaderi, 2024).

Graph-Regularized Multi-Task Learning. Graph-based regularization is a classical approach in multi-task learning (Evgeniou & Pontil, 2004; Maurer, 2006). Bellet et al. extend this to the decentralized setting, where edge weights encode task similarity. Recent surveys by Zhang & Yang (2021) reaffirm the effectiveness of explicit task relationship modeling in improving generalization across heterogeneous clients.

3. Explanation of the Algorithm

Bellet et al. (2018) address the problem of learning *personalized* models across a network of n agents in a fully decentralized and asynchronous fashion. In this setting, each agent i holds a private dataset $S_i = \{(x_i^j, y_i^j)\}_{j=1}^{m_i}$ drawn from its own distribution μ_i. The goal is to learn a distinct model parameter $\theta_i \in \mathbb{R}^p$ for each agent that minimizes local risk while leveraging the similarities between agents to improve generalization.

Because data must remain local and no central coordinator exists, the algorithm relies solely on local computation and message passing along the edges of a communication graph. The method is built upon three main components: (1) a graph-regularized multi-task objective, (2) an asynchronous block coordinate descent update, and (3) a differential privacy mechanism.

3.1. The Graph-Regularized Personalized Objective

Agents are connected through an undirected weighted graph $G = ([n], E, W)$, where the weights $W_{ij} \geq 0$ capture the similarity between the learning tasks of agent i and agent j. Let $D_{ii} = \sum_j W_{ij}$ denote the weighted degree of agent i. To account for data imbalance across the network, the authors introduce a confidence coefficient $c_i \propto m_i$, which scales with the local dataset size.

The formulation proposes a composite objective function $Q(\Theta)$ over the stacked parameter vector $\Theta = (\theta_1^\top, \dots, \theta_n^\top)^\top \in \mathbb{R}^{np}$:

$$\min_{\Theta} Q(\Theta) = \underbrace{\frac{1}{2}\sum_{i<j} W_{ij} \|\theta_i - \theta_j\|^2}_{\text{Similarity of the models}} + \underbrace{\mu \sum_{i=1}^{n} D_{ii} c_i L_i(\theta_i; S_i)}_{\text{Weighted Local Losses}}, \tag{1}$$

where $L_i(\theta_i; S_i) = \frac{1}{m_i}\sum_{j=1}^{m_i} \ell(\theta_i; x_i^j, y_i^j) + \lambda_i \|\theta_i\|_2^2$ is the regularized local empirical loss.

Interpretation of Terms. The first term enforces *smoothness* over the graph, penalizing disagreement between agents that are strongly connected (large W_{ij}). The second term ensures individual accuracy. The hyperparameter $\mu > 0$ controls the trade-off. As $\mu \to \infty$, agents ignore neighbors and perform purely local learning (no collaboration). As $\mu \to 0$, the smoothness term dominates, forcing a global consensus. This objective is strictly convex provided the local losses L_i are strictly convex.

3.2. Asynchronous Decentralized Optimization

To minimize (1) without a central clock, the authors utilize an **Asynchronous Block Coordinate Descent (BCD)** approach. The network time is modeled via independent Poisson clocks (rate 1) for each agent. This is equivalent to a global clock where, at each iteration t, a single agent i is selected uniformly at random to wake up.

3.2.1. The Update Rule

When agent i wakes up at time t, it updates its own parameter θ_i while keeping all other $\theta_{j\neq i}$ fixed. This corresponds to a partial gradient step on the global objective with respect to the i-th block. Using a local block-Lipschitz constant $L_i = D_{ii}(1 + \mu c_i L_i^{loc})$, the update takes the form of a convex combination:

$$\theta_i^{(t+1)} = (1-\alpha_i)\theta_i^{(t)} + \alpha_i \left(\sum_{j \in \mathcal{N}_i} \frac{W_{ij}}{D_{ii}} \theta_j^{(t)} - \mu c_i \nabla L_i(\theta_i^{(t)}) \right), \tag{2}$$

where the step size α_i is defined as:

$$\alpha_i = \frac{1}{1 + \mu c_i L_i^{loc}} \in (0, 1]. \tag{3}$$

Interpretation. The update in Eq. (2) reveals the collaborative mechanism. The new model is a weighted average of: (1) **Inertia** (the agent's previous model), (2) **Collaboration** (the weighted average of neighbors' models), and (3) **Personalization** (a descent step along the local gradient). Crucially, agents with higher confidence c_i (more data) place greater weight on their local gradient, effectively "pulling" their neighbors toward a more accurate solution.

3.3. Differential Privacy Mechanism

In decentralized settings, the sequence of broadcasted models $\theta_i^{(t)}$ can leak information about the private dataset S_i. To prevent this, Bellet et al. integrate a Differentially Private (DP) mechanism directly into the optimization loop.

3.3.1. Noisy Gradient Updates

The authors employ *gradient perturbation*. Before performing the update in Eq. (2), agent i adds noise to its local

gradient computation. The private update becomes:

$$\theta_i^{(t+1)} = (1-\alpha_i)\theta_i^{(t)} + \alpha_i\left(\sum_{j\in\mathcal{N}_i}\frac{W_{ij}}{D_{ii}}\theta_j^{(t)} - \mu c_i\left(\nabla L_i(\theta_i^{(t)}) + \boldsymbol{\eta}^{(t)}\right)\right). \tag{4}$$

where $\boldsymbol{\eta}^{(t)} \sim \text{Laplace}(0, s_i^{(t)})^p$ is a noise vector drawn from a Laplace distribution.

3.3.2. Sensitivity and Noise Calibration

To satisfy ϵ-DP, the scale of the noise $s_i^{(t)}$ depends on the sensitivity of the gradient function. Assuming the loss function is L_0-Lipschitz with respect to the L_1-norm, changing one data point in S_i changes the local gradient term by at most $\Delta = \frac{2L_0}{m_i}$. Consequently, the required noise scale is inversely proportional to the local dataset size m_i. This highlights a key advantage of the framework: agents with more data add less noise, thereby maintaining higher utility while preserving the same level of privacy.

3.3.3. Composition and Privacy Budget

Since agents communicate iteratively, privacy loss accumulates over time. The algorithm utilizes the advanced composition results for differential privacy (Kairouz et al., 2015) to track the cumulative privacy budget (ϵ, δ). The algorithm terminates for agent i when its specific privacy budget is exhausted.

3.4. Convergence and Analysis

The theoretical analysis provided by Bellet et al. covers both the convergence rate and the utility-privacy trade-off.

Convergence Rate. Under the assumption that the local losses are strongly convex and smooth, the non-private version of the algorithm converges *linearly* in expectation to the global optimum Θ^*. Specifically, for a strongly convex objective with parameter σ, the expected error at iteration T is bounded by:

$$\mathbb{E}[Q(\Theta^{(T)}) - Q(\Theta^*)] \leq \left(1 - \frac{\sigma}{nL_{\max}}\right)^T (Q(\Theta^{(0)}) - Q(\Theta^*)),$$

where $L_{\max}$ is the maximum block Lipschitz constant.

Utility under Privacy. With the addition of privacy noise $\boldsymbol{\eta}$, exact convergence to Θ^* is no longer possible. Instead, the algorithm converges to a neighborhood of the optimal solution. The error bound decomposes into two terms:

1. **Optimization Error:** Decays geometrically over time (as in the non-private case).
2. **Privacy Noise Floor:** An accumulated error term proportional to the variance of the injected noise.

This creates a trade-off: running the algorithm longer reduces optimization error but requires adding more noise (or consuming more budget), which eventually increases the noise floor. There exists an optimal stopping time that minimizes the total excess risk.

4. Analysis and Extensions

4.1. Beyond Convexity

The convergence analysis in Bellet et al. (2018) relies heavily on the assumption that the local loss functions L_i are strongly convex and smooth. This assumption guarantees the linear convergence rate derived in Proposition 1. However, modern machine learning applications, particularly deep learning, involve highly non-convex objectives. To bridge this gap, we formalize the convergence guarantee for non-convex smooth functions.

4.1.1. Theoretical justification:

We verify that the algorithm satisfies the necessary conditions: (1) the global objective $\mathcal{Q}$ is L-smooth; (2) the privacy noise acts as an unbiased stochastic gradient estimator with bounded variance σ^2; and (3) the Poisson clock update schedule ensures every agent updates with non-zero probability. Under these conditions, we state the following theorem (proof provided in Appendix A).

Theorem: Let $\{\Theta^{(k)}\}$ be the sequence of iterates generated by the decentralized algorithm. If the step size is chosen as $\alpha = \frac{1}{\sqrt{K}}$, where K is the total number of iterations, then the expected gradient norm satisfies:

$$\min_{0\leq k<K} \mathbb{E}\|\nabla\mathcal{Q}(\Theta^{(k)})\|^2 \leq \frac{2n(\mathcal{Q}(\Theta^{(0)}) - \mathcal{Q}^*)}{\sqrt{K}} + \frac{nL\sigma^2}{\sqrt{K}}. \tag{5}$$

This result confirms that the algorithm converges to a stationary point at a rate of $\mathcal{O}(1/\sqrt{K})$. It can also be applied to stochastic gradient algorithms like SGD (which are ubiquitous in Deep Learning), as long as the stochastic gradient estimator is unbiased and with bounded variance.
A comparable result has been proven for Asynchronous Block Coordinate Descent in the presence of bounded delays (Zhou & Pu, 2025), providing a rigorous foundation for scenarios where the zero-delay assumption of Bellet et al. (2018) does not hold.

4.1.2. Numerical Experiment

To empirically validate our theoretical extension to non-convex objectives, we simulate a decentralized learning task on the MNIST dataset using a network of 20 agents arranged

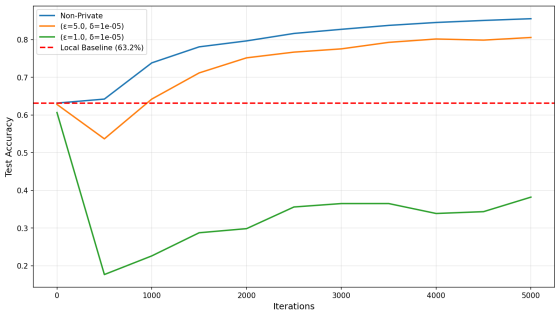


Figure 1. Decentralized MNIST Classification results

in a Watts-Strogatz small-world topology ($k = 5, p = 0.3$). Each agent possesses a small local partition of 50 samples and trains a non-convex Multi-Layer Perceptron (MLP) with one hidden layer (64 hidden neurons). We set the similarity weights $W_{ij} = 1$ as the task of the agents are very similar. The local models are first trained locally (warm start), and then trained collaboratively using the Asynchronous Block Coordinate Descent algorithm described by (Bellet et al., 2018).

Results: The algorithm demonstrates convergence, with a significant gain in test accuracy. Agents starting with a baseline accuracy of approximately 63% (trained on local data only) achieved a test accuracy of around 85% in the non-private case, through collaboration. This suggests that the pairwise graph regularization effectively transfers knowledge even in non-convex optimization landscapes. In non-convex landscapes like neural networks, training on such small datasets typically leads to severe overfitting and poor generalization, as agents get trapped in sharp, low-quality local minima. The graph regularization term $\frac{1}{2}\sum W_{ij}\|\theta_i - \theta_j\|^2$ seems to effectively pull agents out of these sharp local minima towards a consensus region that generalizes better.

However, the differentially private models degrades rapidly upon the injection of noise. In the non-convex setting, the optimization process appears highly sensitive to the variance introduced by the privacy mechanism, leading to a substantial drop in utility even at moderate privacy budgets. This suggests that while non-convex collaboration is powerful, the resulting optimization trajectory is far less robust to privacy noise than in the convex case. This aligns with our derived bound in the theorem , where the error bound is proportional to the noise variance σ^2. With a convex loss function, a noisy gradient step might oscillate but will generally remain directed towards the minimum. In a non-convex landscape with many saddle points and ridges, a noisy update can easily push the parameters into a different, less optimal local minimum or saddle point, from which the algorithm cannot recover.

4.2. Similarity Weights and Graph Topology

The algorithm assumes the existence of a fixed weight matrix W where W_{ij} reflects the "task relatedness" between agents i and j. In the original experiments, these weights are derived from ground-truth similarities (e.g., angles between target models) which are unknown in practice. While the authors suggest that these weights can be derived from user profiles or computed privately from local datasets, the strategies they mention rely on heavy cryptographic operations and strict synchronization, which goes against the asynchronous and simple nature of the problem.

Sensitivity to Weight Accuracy. We first demonstrate that accurate knowledge of W_{ij} is not merely beneficial but essential for the algorithm's success. Theoretical analysis suggests that if W_{ij} is high between agents with dissimilar tasks (negative transfer), the regularization term $\sum W_{ij}\|\theta_i - \theta_j\|^2$ forces their models to converge towards a consensus that is suboptimal for both. This "pollution" effect can degrade local performance below the baseline of isolated learning. Consequently, in the absence of reliable metadata to estimate task similarity, using an ill-defined or random weight matrix risks actively harming utility.

To investigate practical alternatives to oracle weights, we conducted experiments using "blind" topologies where no task similarity information is assumed (i.e., uniform weights). We first replicated the synthetic linear classification experiment from Bellet et al. (2018) ($N = 100$) to establish a valid baseline. We then modified the communication graph to test three distinct topologies with random agent placement: a complete graph, a cycle graph and a 2D grid. As expected, the test accuracy of the models trained with those agnostic graphs deteriorates very quickly.

Dynamic Weight Learning. Given the limitations of static or random topologies, we attempted to estimate the weights W_{ij} dynamically during training. We implemented a simple heuristic where agents periodically update their neighbor weights based on the cosine similarity of their current model parameters: $W_{ij} \propto \langle \tilde{\theta}_i, \tilde{\theta}_j \rangle$. However, our experiments showed that this approach fails in the private setting. The combination of differential privacy noise and the stochasticity of early-stage training renders the similarity metric unreliable, leading to erratic topology changes that hinder convergence rather than helping it.A more principled solution to this problem is proposed by Zantedeschi et al. (2020). In their work on *Fully Decentralized Joint Learning*, they formulate the task as a bi-level optimization problem, where the graph topology and the personalized models are learned simultaneously. Adopting such a framework could potentially allow the network to discover the underlying task structure without relying on unstable heuristics or "oracle" knowledge.

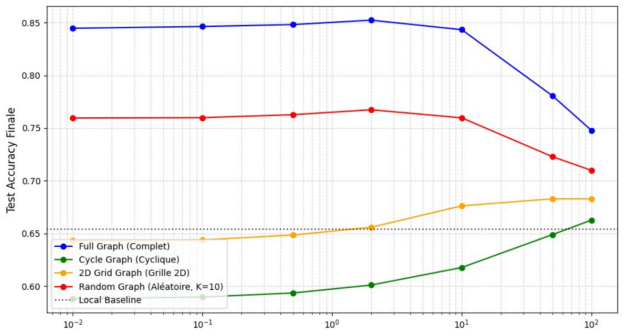


Figure 2. Impact of graph topology - Test accuracy with regard to μ (log scale)

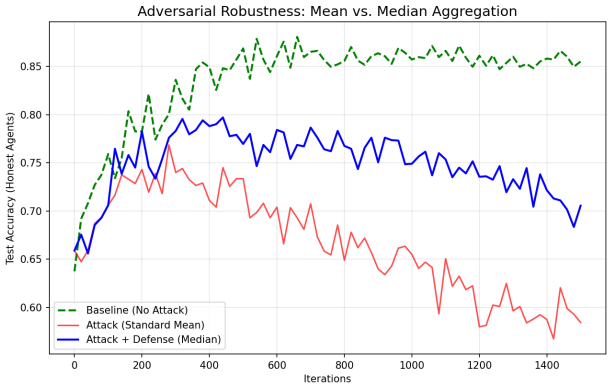


Figure 3. Mean Aggregation vs Median Aggregation

Impact of Graph Topology. Even when the weights W_{ij} accurately reflect task similarities, the structure of the communication graph fundamentally dictates the algorithm's ability to leverage collaborative information. In a real scenario, the topology could be dictated by external hard constraints. Our experiments reveal a clear hierarchy in performance across different topologies. The **Complete Graph** consistently outperforms sparse structures, as its high connectivity allows agents to directly access information from their most relevant peers, maximizing the benefit of the graph regularization term. In contrast, constrained topologies like the **2D Grid** and **Cyclic Graph** suffer from structural bottlenecks; agents are forced to communicate only with immediate geometric neighbors rather than their most task-similar peers, which limits the effectiveness of knowledge transfer regardless of the regularization strength. This behavior is modulated by the parameter μ: at low values of μ, the algorithm prioritizes consensus, which is effective in the Complete graph but detrimental in sparse graphs where agents may be pulled towards irrelevant neighbors. As μ increases, prioritizing local accuracy, performance across all topologies converges closer to the purely local baseline, confirming that the topology's primary role is to facilitate the "collaborative boost" available in the lower-μ regime.

4.3. Adversarial Robustness

The framework proposed by Bellet et al. (2018) relies on the *honest-but-curious* threat model. In this model, agents are assumed to truthfully follow the protocol, performing correct gradient updates and broadcasting true parameters, but may inspect received messages to infer private information. While this assumption simplifies the theoretical analysis, it is often unrealistic in open peer-to-peer networks. Real-world deployments must contend with Byzantine or malicious agents who actively attempt to sabotage the learning process.

To demonstrate the fragility of the standard algorithm under such threats, we conducted an adversarial experiment using the previous synthetic linear classification task with $N = 100$ agents. We introduced a significant fraction of malicious agents (35%) into the network. Instead of learning, these adversaries broadcast high-variance random noise ($\theta_{mal} \sim \mathcal{N}(0, 25)$) to their neighbors at every iteration.

We compared two aggregation strategies:

1. **Standard Mean Aggregation:** The original update rule, which computes a weighted average of neighbors' models.
2. **Robust Median Aggregation:** A proposed defense where the weighted average is replaced by a coordinate-wise median, a statistic known for its high breakdown point (robustness to outliers).

Results: As illustrated in Figure 3, the standard algorithm (Mean Aggregation) is catastrophic under attack; the test accuracy of honest agents collapses to near-random performance as the malicious noise propagates through the linear averaging. In contrast, the robust method manages to recovers an accuracy greater than the local baseline (around 0.7), proving that replacing the sensitive mean operator is essential for deploying decentralized learning in untrusted environments.

Limitations and Future Defenses. While the median aggregator works well for simple linear models, it has significant limitations for deep learning. Since it calculates the median for each parameter independently, it can break the internal structure of the update vector in high-dimensional neural networks. Furthermore, sophisticated attackers can still trick simple statistical filters. For more sophisticated cases, one could use distance-based methods like Krum. Unlike the median, Krum selects a single, complete update from a neighbor that is closest to the majority. This

approach preserves the coherence of the model updates, though it requires more computing power to calculate the distances between all neighbors.

5. Conclusion

This paper presented a critical analysis and extension of the personalized peer-to-peer learning framework introduced by Bellet et al. (2018). While the original work provides a solid theoretical foundation for decentralized coordinate descent, our investigation highlights significant challenges in translating these guarantees to realistic settings.

First, we demonstrated that the framework can be extended to non-convex objectives, achieving substantial utility gains in data-scarce neural network training. However, we also identified a "privacy cliff": the utility of non-convex models degrades much more rapidly under differential privacy than their convex counterparts. Second, we underlined the importance of the choice of the similarity weights and the impact of graph topology. Finally, we exposed the fragility of the honest-but-curious assumption, proving that simple robust aggregation mechanisms (like the median) are strictly necessary to prevent system collapse in the presence of Byzantine adversaries. Future work could focus on integrating bi-level optimization for dynamic graph learning and developing computationally efficient robust aggregators suitable for high-dimensional deep learning updates.

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A. Convergence Proof for Non-Convex Objectives

In this appendix, we provide the detailed proof for Theorem 4.1.1, demonstrating the convergence of the algorithm in Bellet et al. (2018) under non-convexity.

A.1. Setup and Assumptions

We consider the global objective $\mathcal{Q}(\Theta)$ defined in Eq. (1). We assume:

1. **Smoothness:** $\mathcal{Q}$ is L-smooth, i.e., $\|\nabla\mathcal{Q}(\Theta) - \nabla\mathcal{Q}(\Theta')\| \leq L\|\Theta - \Theta'\|$.
2. **Bounded Variance:** The privacy noise η is zero-mean and has bounded variance $\mathbb{E}[\|\eta\|^2] \leq \sigma^2$.
3. **Poisson Update:** At each step k, a single agent i_k is selected uniformly at random (prob. $1/n$).

A.2. Proof Derivation

From the L-smoothness of $\mathcal{Q}$, for any update $\Delta^{(k)} = \Theta^{(k+1)} - \Theta^{(k)}$, we have:

$$\mathcal{Q}(\Theta^{(k+1)}) \leq \mathcal{Q}(\Theta^{(k)}) + \langle\nabla\mathcal{Q}(\Theta^{(k)}), \Delta^{(k)}\rangle + \frac{L}{2}\|\Delta^{(k)}\|^2. \tag{6}$$

The update only affects block i_k: $\Delta^{(k)}_{i_k} = -\alpha(\nabla_{i_k}\mathcal{Q}(\Theta^{(k)}) + \eta^{(k)})$. We take the expectation $\mathbb{E}_k$ over the random choice of agent i_k and the noise $\eta^{(k)}$.

The linear term simplifies as follows, noting that $\mathbb{E}_{i_k}[\nabla_{i_k}\mathcal{Q}] = \frac{1}{n}\nabla\mathcal{Q}$:

$$\mathbb{E}_k[\langle\nabla\mathcal{Q}, \Delta^{(k)}\rangle] = -\alpha\sum_{i=1}^{n}\frac{1}{n}\mathbb{E}_\eta\langle\nabla_i\mathcal{Q}, \nabla_i\mathcal{Q} + \eta\rangle \tag{7}$$

$$= -\frac{\alpha}{n}\|\nabla\mathcal{Q}(\Theta^{(k)})\|^2. \tag{8}$$

The quadratic term is bounded by separating the gradient and noise variance:

$$\mathbb{E}_k[\|\Delta^{(k)}\|^2] = \alpha^2\sum_{i=1}^{n}\frac{1}{n}\mathbb{E}_\eta\|\nabla_i\mathcal{Q} + \eta\|^2 \tag{9}$$

$$= \frac{\alpha^2}{n}\|\nabla\mathcal{Q}(\Theta^{(k)})\|^2 + \alpha^2\sigma^2. \tag{10}$$

Substituting back into the smoothness inequality:

$$\mathbb{E}_k[\mathcal{Q}(\Theta^{(k+1)})] \leq \mathcal{Q}(\Theta^{(k)}) - \frac{\alpha}{n}\left(1 - \frac{L\alpha}{2}\right)\|\nabla\mathcal{Q}(\Theta^{(k)})\|^2 + \frac{L\alpha^2\sigma^2}{2}. \tag{11}$$

Assuming $\alpha \leq 1/L$, we have $(1 - L\alpha/2) \geq 1/2$. Rearranging terms:

$$\frac{\alpha}{2n}\|\nabla\mathcal{Q}(\Theta^{(k)})\|^2 \leq \mathcal{Q}(\Theta^{(k)}) - \mathbb{E}_k[\mathcal{Q}(\Theta^{(k+1)})] + \frac{L\alpha^2\sigma^2}{2}. \tag{12}$$

Summing over $k = 0, \ldots, K-1$ and dividing by K:

$$\frac{1}{K}\sum_{k=0}^{K-1}\mathbb{E}\|\nabla\mathcal{Q}(\Theta^{(k)})\|^2 \leq \frac{2n(\mathcal{Q}(\Theta^{(0)}) - \mathcal{Q}^*)}{K\alpha} + Ln\alpha\sigma^2. \tag{13}$$

Substituting $\alpha = 1/\sqrt{K}$ yields the rate $\mathcal{O}(1/\sqrt{K})$. □

B. Code and Implementation Details

Repository Link: https://github.com/alexandrejiao/EA_collaborative_learning